

(a)

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In [7]: import numpy as np
        from scipy.interpolate import lagrange
        import matplotlib.pyplot as plt

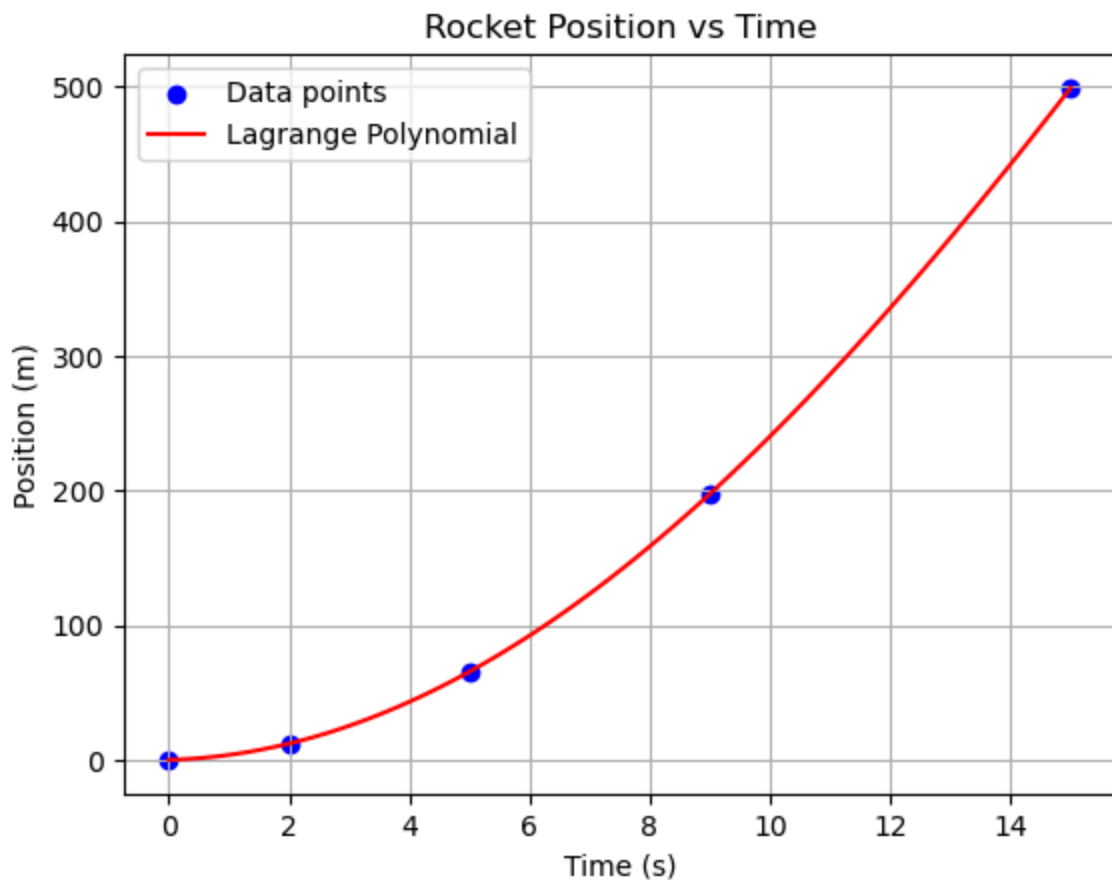
        t = np.array([0, 2, 5, 9, 15])
        h = np.array([0.0, 12.1, 65.5, 197.6, 499.0])

        poly = lagrange(t, h)

        t_fit = np.linspace(0, 15, 100)
        h_fit = poly(t_fit)

        plt.scatter(t, h, color='blue', label='Data points')
        plt.plot(t_fit, h_fit, color='red', label='Lagrange Polynomial')
        plt.xlabel('Time (s)')
        plt.ylabel('Position (m)')
        plt.title('Rocket Position vs Time')
        plt.legend()
        plt.grid(True)
        plt.show()

        print("Lagrange polynomial coefficients:", poly.coef)
```



Lagrange polynomial coefficients: [-1.03276353e-03 -2.92022792e-03 2.41071937e+00
1.24850427e+00
0.00000000e+00]

(b)

```
In [8]: import numpy as np
        from scipy.interpolate import lagrange
        import matplotlib.pyplot as plt

        t = np.array([0, 2, 5, 9, 15])
        h = np.array([0.0, 12.1, 65.5, 197.6, 499.0])

        poly = lagrange(t, h)

        # Velocity  $v(t) = d/dt [h(t)]$ 
        velocity_poly = np.polyder(poly)

        t_1 = 7
        velocity_at_7 = velocity_poly(t_1)

        print(f"\nVelocity at t = {t_1} seconds: {velocity_at_7:.4f} m/s")
```

Velocity at t = 7 seconds: 33.1524 m/s

In []: